



HK IMO Training 2021-2022
Problem Set 13
For 8 January 2022



Problem 49. Find all nonzero monic polynomials $P(x)$ with integer coefficients such that its roots are exactly $P(0), P(1), \dots, P(n-1)$ (counted with multiplicity) for some positive integer n .

Problem 50. Prove that there exists a positive integer n not exceeding 1000000 satisfying the following property: the decimal representation of 5^n has 6 consecutive zeros.

Problem 51. In $\triangle ABC$, let D, E, F be the contact points of the incircle with the sides BC, CA, AB respectively. Let Γ_1 be a circle passing through E and F , Γ_2 be a circle passing through F and D , Γ_3 be a circle passing through D and E , such that Γ_2 and Γ_3 meet again on the line AD , Γ_3 and Γ_1 meet again on the line BE , Γ_1 and Γ_2 meet again on the line CF . Suppose Γ_1 intersects the segments DE and DF at A_1 and A_2 respectively, Γ_2 intersects the segments EF and ED at B_1 and B_2 respectively, Γ_3 intersects the segments FD and FE at C_1 and C_2 respectively. Prove that $A_1, A_2, B_1, B_2, C_1, C_2$ are concyclic.

Problem 52. The king is placed in one of the cells of an $n \times n$ chessboard, where n is a positive integer. In each move, the king can move to an adjacent cell sharing at least one common vertex. Consider all possible paths of the king such that each cell is visited exactly once (the initial cell should not be visited again). What is the largest possible number of moves that the king moves diagonally?